CYTOTOXICITY ASSAY AND DNA FRAGMENTATION

Cytotoxicity Assay (MTT Assay)

HeLa cells were seeded into 96-well plates at a density of approximately 1×10^4 cells per well and incubated for 24 h. Cells were treated with different concentrations of *Acalypha indica* extract, while untreated cells served as control. After 24 h, MTT solution was added and incubated for 4 h. Formazan crystals formed were dissolved in DMSO, and absorbance was measured at 570 nm. Cell viability (%) was calculated using:

Cell viability (%) = (Sample OD/ Control OD)×100

DNA Fragmentation Analysis

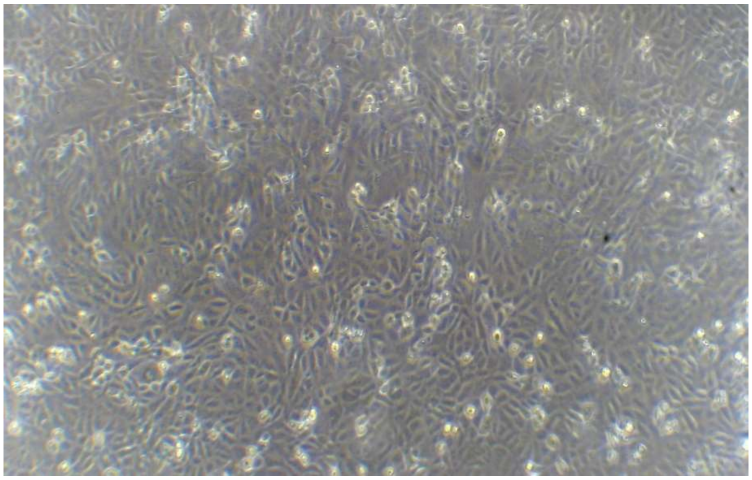
HeLa cells were treated with extract at IC_{50} concentration and incubated for 24 h. Cells were harvested, washed with PBS, and subjected to DNA isolation. Extracted DNA was analyzed using 1% agarose gel electrophoresis containing ethidium bromide. DNA ladder was used as a marker. DNA bands were visualized under UV light, and the presence of a ladder pattern indicated apoptosis.

RESULTS

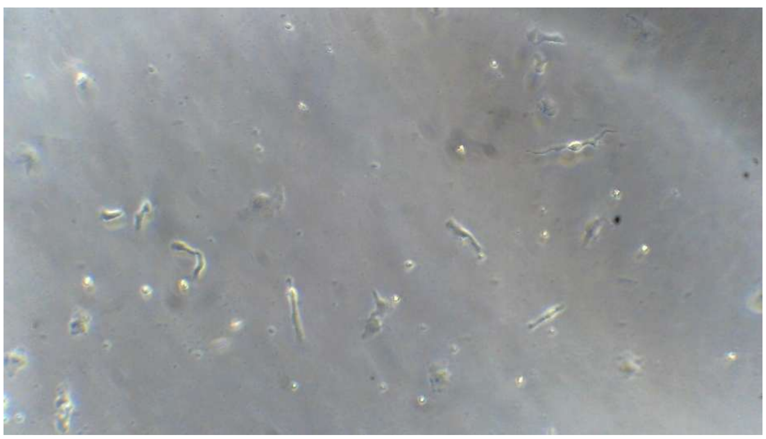
CYTOTOXICITY ASSAY:

Concentration of (ug/ml)	% of Cell Viability (Triplicate Values)				% of viability
	Trial 1	Trial 2	Trial 3	Average	
250	0.047	0.053	0.042	0.047333	48.2%
125	0.064	0.071	0.067	0.067333	68.7%
62.5	0.073	0.075	0.074	0.067333	68.7%
31.25	0.081	0.083	0.084	0.082667	84.3%
15.6	0.089	0.087	0.092	0.089333	91.1%
CONTROL	0.097	0.098	0.099	0.098	100%

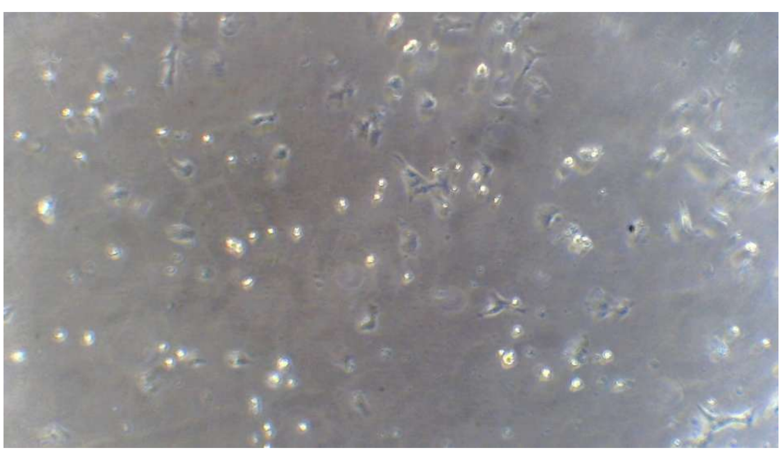
Control



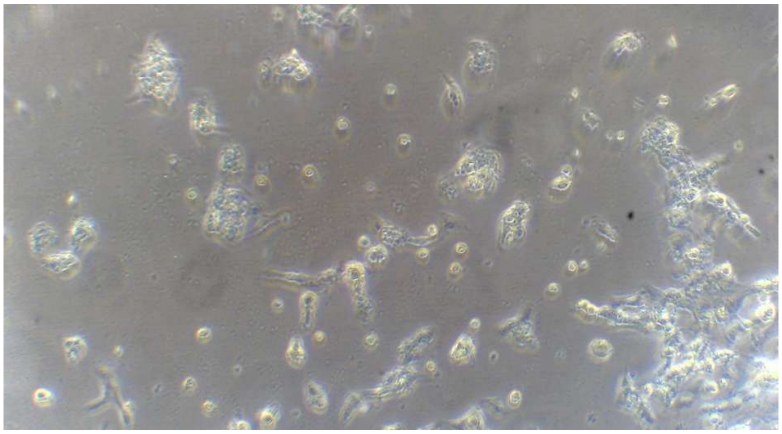
250



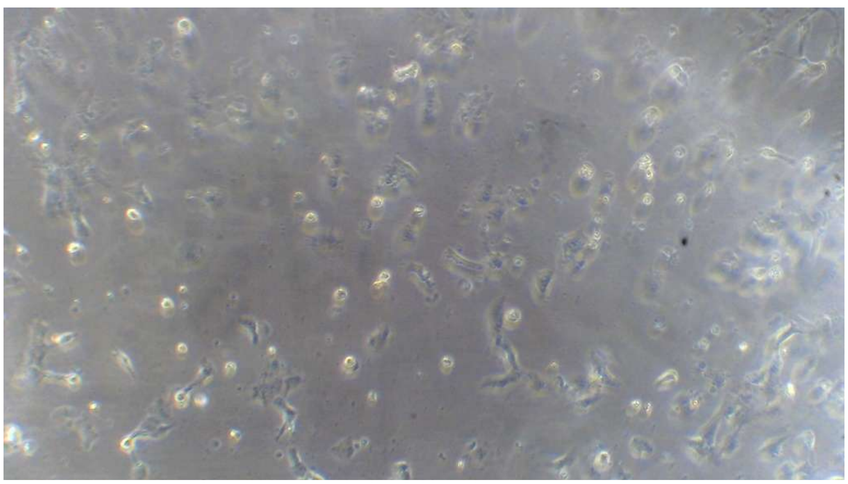
125



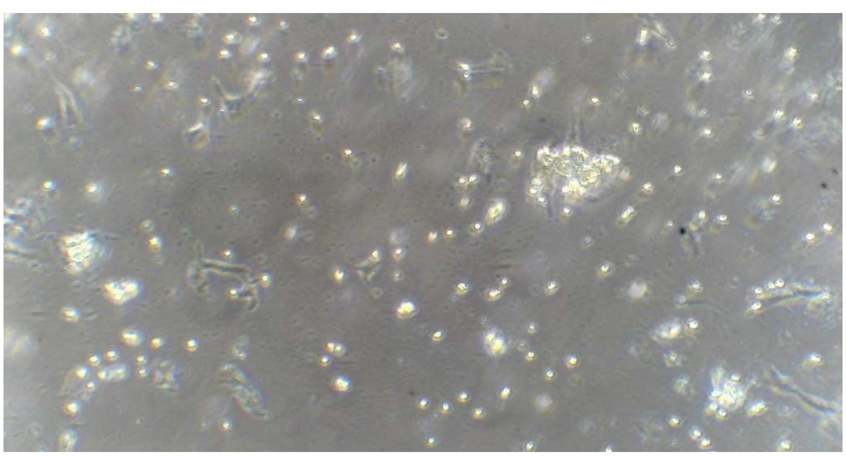
62.5



31.25

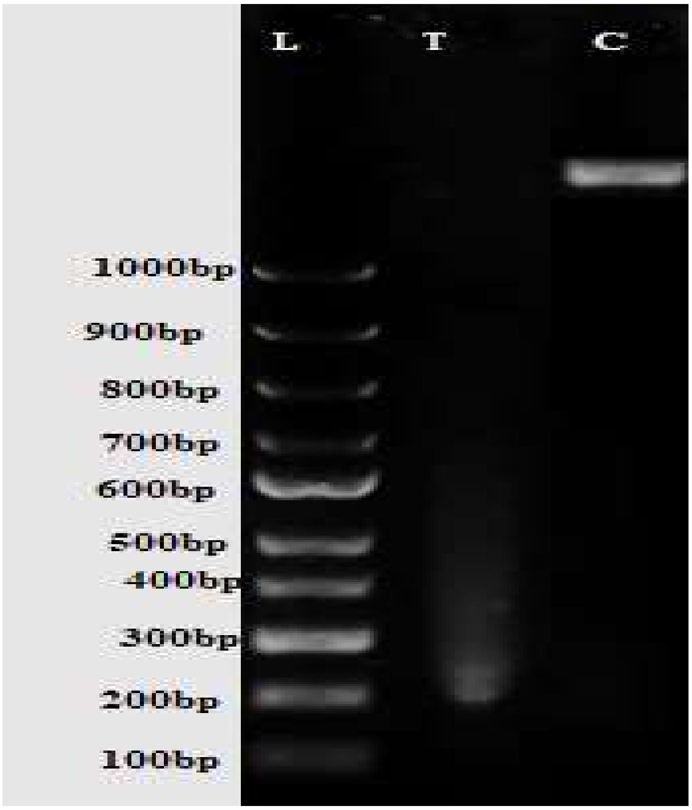


15.6



This shows dose- dependent decrease in cell viability

DNA FRAGMENTATION ANALYSIS



Qualitatively, DNA fragmentation was observed to be very high > 90% in the treated sample (T)